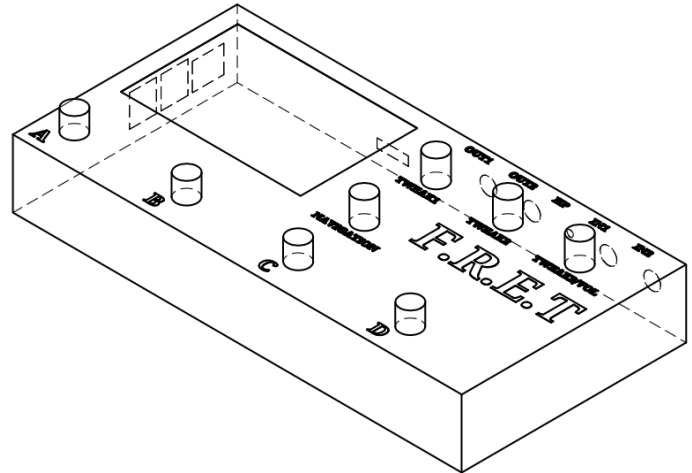




F.R.E.T. - Flexible Real-Time Effects Terminal

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Motivation and Background

Guitarists rely on multiple effects pedals to produce their own unique sound. While individual pedals can give guitarists a large assortment of effects, using multiple pedals can get messy quickly by increasing costs, requiring additional power and cables, and making things more difficult to change between effects. With the power of digital signal processing found in our project, many effects can be combined into a single programmable system, all inside of a guitar pedal interface.

A main drawback of these digital multi-effects pedals is that they crash and have poor tone at high volumes. Professionals steer clear because of this, despite their versatility and ability to produce an extremely wide array of sounds. This is due to overheating, high CPU utilization, and poor separation between analog signals and noise-producing components, and more. A better design for such kits would be to address one or more of these issues, improving the product by making it stage-ready, or closer to it.

Electric guitars produce an analog electrical signal that is modified by the user by implementing analog or digital effects pedals before being sent to an amplifier. Effects like distortion, delay, chorus, and reverb modify the contents of the guitar's analog electrical signal including amplitude, frequency, and phase. When using a digital effects system, the analog signal coming from the guitar must be converted into digital samples. These samples are then processed using algorithms before being converted back into an analog signal to be sent to an amplifier. This process is the fundamental design background for our project.

Our proposed system will seek to use a high-quality digital signal processor. The guitar signal will enter the pedal through a standard cable and pass through an analog input stage to condition the high impedance. An audio codec will perform the analog-to-digital conversion so that the digital signal processor can process the signal. Once the signal is processed, the audio codec will convert the digital signal back to analog. Finally, an output conditioning stage will ensure that the signal can be transferred to the amplifier. An additional microcontroller will also be used to manage most of the system's controls and peripherals, including potentiometers, encoders, and footswitches. A microcontroller is important to the project because it will allow the guitarist to interact with the pedal without needing a keyboard or mouse.

All these subsystems will be integrated through a custom PCB and will contain the microcontroller, power system, peripherals, and ADC/ACD. This will satisfy the design requirement of a significant custom PCB involving an MCU, power supply unit, and MCU peripherals. As per design instructions, pre-established reference circuits will be used for functions like voltage regulation, ADC/ACD, filters, etc.

The motivation for our project is to develop a programmable digital multi-effects guitar pedal that will combine audio processing, physical control peripherals, and real-time audio-visual feedback into a single user-friendly device. The pedal will allow the user to apply and modify effects like distortion, delay, chorus, equalization, and many other digital effects without requiring a separate physical pedal for each effect. A fundamental feature will be presets, which allow combinations of effects and settings to be stored and recalled, which will allow the user to quickly retrieve their desired sound while playing.

This engineering project will be our team's opportunity to combine several areas of electrical and computer engineering that we have learned during our college careers into a single design.

Goals and Objectives

Goals

Basic Goals:

- Provide user with multiple guitar effects in a single device.
- Allow the user to control and adjust effects while playing the guitar.
- Allow the pedal to operate following an electric guitar and guitar amplifier.
- Provide reliable and safe operation from a single external power source.
- Create a self-contained and portable pedal suitable for normal guitar use.

Advanced Goals:

- Provide real-time visual feedback that represents the guitar signal.
- Provide the user with clear information about the current effects and settings.
- Allow multiple effects to operate together to create more complex sounds.
- Provide a streamlined user experience.

Stretch Goals:

- Allow the user to create and quickly recall customized sounds using presets.
- Allow the user to view how their enabled effects change the sound.
- Allow users to control and configure the pedal wirelessly.
- Provide additional useful tools to a guitarist, such as tuning and tempo control.
- Provide additional experimental guitar effects.
- Provide information about the operating condition of the pedal for troubleshooting.
- Allow the user to record data.
- Allow external devices to control the pedal.
- Improve system reliability through automatic fault detection and recovery.

Objectives

Basic Objectives:

- Design and implement a real-time digital audio processing system capable of applying multiple guitar effects.
- Implement at least five digital guitar effects, including distortion, delay, chorus, equalization, and reverb.
- Design an analog input and output signal path capable of interfacing the system with a standard electric guitar and guitar amplifier.
- Integrate an audio codec to perform ADC and DAC at a minimum sampling rate of 45 kHz.
- Maintain an input-to-output audio processing latency of 10 ms or less.
- Integrate physical footswitches and rotary controls that allow the user to enable effects and adjust effect parameters.
- Design a power subsystem capable of supplying all required system voltage rails from a single external DC power source.

- Design and manufacture a custom PCB containing the MCU, power circuitry, and peripherals.
- Integrate the audio processing hardware, control electronics, user controls, connectors, and power system into a single portable enclosure.

Advanced Objectives:

- Implement a graphical user interface that displays active effects, parameter values, and system information.
- Support the simultaneous processing of at least three digital guitar effects.
- Develop an effects-processing chain that allows individual effects to be enabled, disabled, and adjusted without interrupting the audio signal.
- Implement responsive interaction between the display, physical controls, and MCU.
- Maintain a user-interface response time of 100 ms or less for effect and parameter changes.

Stretch Objectives:

- Implement a preset system capable of storing and recalling at least 10 user-defined effect configurations.
- Display the analyzed guitar signal as a digital spectrum visualizer with at least 16 frequency bands and a refresh rate of at least 20 updates per second.
- Implement before-and-after signal analysis to visually demonstrate how enabled effects modify the frequency content of the guitar signal.
- Implement Bluetooth communication for wireless configuration and control of the pedal.
- Implement a digital guitar tuner that identifies the detected musical note.
- Develop additional experimental digital effects such as bit crushing, reverse delay, auto-wah, or stutter effects.
- Monitor system operating information such as supply voltage, current consumption, processor temperature, or processing load.
- Implement system fault detection capable of identifying a loss of communication or processing failure and placing the pedal into a safe operating or bypass state.

Overall Marketing Goals:

Sound Quality, ensuring the user and audience can enjoy the sound without distraction.

Versatility, allowing the user to enjoy more features and sounds for less money and setup.

Stable Operation, allowing the end user to rely on our product for use in front of an audience and enjoy the experience our product can provide.

Cost effectiveness, providing the above goals for a reasonable price.

Size, allowing the end user to conveniently employ our product without impedance.

Features/Functionalities

Audio Input and Output

- Standard ¼-inch TS guitar input and amplifier output
- High-impedance guitar input stage
- Input buffering and anti-aliasing filter
- Audio codec for ADC and DAC
- Minimum 45 kHz audio sampling rate
- 24-bit audio conversion
- Digital audio communication between codec and MCU
- Output reconstruction/filtering stage
- Low-impedance buffered output

Digital Effects Processing

- Real-time DSP performed by the MCU
- Implement common effects such as compression, distortion/overdrive, delay, chorus, reverb, equalization, etc.
- Adjustable effect parameters
- Ability to enable and disable individual effects

Peripherals

- Multiple footswitches for effect control
- Rotary encoders for parameter adjustment
- Additional push buttons as needed
- Status LEDs
- Optional expression-pedal input
- Optional voltage/current monitoring
- Debugging and programming interface

User Interface

- 4.3-inch IPS TFT display
- 800 × 480 resolution
- Effect selection and configuration menus
- Display of currently enabled effects
- Display of effect parameter values
- Graphical interface for effect configuration
- System status and diagnostic information

Power System

- Single external DC power input
- Input overcurrent protection
- Reverse-polarity protection
- Regulated 3.3 V supply
- Additional regulated voltage rails as required
- Switching regulators for efficient power conversion
- Power-status indication
- Voltage test points for troubleshooting

- Supply at least 125% of calculated maximum load capacity

PCB and Enclosure

- Custom PCB containing the MCU
- Audio codec integrated onto the custom PCB
- Input and output analog circuitry
- Integrated voltage regulators and power protection
- Interfaces for peripherals
- PCB-mounted or panel-mounted 1/4-inch audio connectors
- Accessible footswitches and rotary controls
- Portable enclosure suitable for normal guitar use

Preset System

- Store user-created effect configurations
- Recall saved configurations
- Store at least 5 presets
- Save enabled/disabled state of each effect
- Save individual effect parameter values
- Preset selection through physical controls or touchscreen
- Display currently selected preset
- Nonvolatile storage so presets remain after power is removed

Multiple Simultaneous Effects

- Run at least three digital effects simultaneously
- User-selectable combination of active effects
- Continuous real-time audio processing while multiple effects are active
- Individual parameter adjustment for each active effect
- Enable or disable an effect without stopping audio processing
- Maintain acceptable input-to-output latency with multiple effects active
- Allow effects to be arranged into a digital signal-processing chain
- Display all currently active effects on the user interface

Existing Related Projects

Pi-Stomp V3

An open-source, DIY multi-effects pedal kit built around a Raspberry Pi 5. It runs a full Linux pro-audio stack, mod-host, and MOD-UI, hosting hundreds of LV2 effects and amp-modeling plugins, including Neural Amp Modeler support. Physical controls consist of four footswitches and a set of rotary encoders, paired with a small onboard color display for navigating presets. This pedal is popular among DIY pedal enthusiasts due to community documentation and support. Users can install community-made effects or write their own. It sells as a \$299 kit and represents the "affordable, hackable" end of the multi-effects spectrum. Despite being made by a small company, Tree Fall Sound, the Pi-Stomp has amazing sound quality. The pedal includes a codec connected to the Raspberry Pi, which supports a sampling rate of 96kHz. Known issues regarding the pedal include slow boot time, software delays at high frequency, and overheating, which can cause the system to crash. [A.1]



Figure 1. Pi-Stomp V3 guitar pedal

Sonicake Matribox

A budget-tier, closed-hardware multi-effects processor from Sonicake, with a lineup spanning around \$150-350 depending on tier (base Matribox, Matribox II, and the touchscreen Matribox II Pro). The current Matribox II (\$229.99) offers 63 amp models, 93 effect/pedal simulations, and 61 cabinet simulations, running up to 10 simultaneous, reorderable effect modules on a color LCD screen. On the lower tier, processing is capped at 24-bit/44.1kHz, which is lower than competitors who aim for 96kHz. Provided with only three footswitches, and no XLR output, the lower tier models can't offer users a steady gigging experience. The same core signal-processing remains in higher tier models such as the Matribox II Pro. The differences include additional footswitches, XLR outputs, and a full color touchscreen. [A.2]



Figure 2. Sonicake Matribox guitar pedal

FunBox

An open-source, nine single-effect guitar pedal platform, built around the Daisy Seed chip by Electro-Smith. The MCU on the Daisy Seed is the ARM Cortex-M7 (STM32H750), running at 480MHz. The board is equipped with stereo audio IQ, 24-bit/96kHz audio hardware. This is more focused on the DIY aspect of pedal building. For the price range, the FunBox comes in at the lowest, which is under \$100. There are flaws in the system such as having no signal chain (each individual effect pedal needs a physical cable to connect), re-flashing firmware required to switch effect, no bypass relay, and no display screen. [A.3]



Figure 3. Funbox guitar pedal

Engineering Specifications

Engineering specifications refer to the quantitative measurement and requirements for both the system as a whole (Table 1), and the individual components contained within (Table 2). These values are determined by assessing the marketing and engineering goals, then determining what quantitative measurements may achieve those goals.

<i>Overall System Engineering Specifications</i>			
#	Engineering Specification	Quantitative Measure	Verification Method
1	Audio sampling rate	≥ 45 kHz	Verify the codec and software sampling configuration
2	Audio processing latency	≤ 10 ms guitar input - $>$ processed audio output	Apply input waveform and measure the time difference between input and output with oscilloscope
3	<u>Startup time</u>	≤ 30 s from power-on to operational audio processing	Measure from power application to operation
4	User interface response time	≤ 100 ms from user input to displayed parameter change	Measure the time between control inputs and screen response
5	UI refresh rate	≥ 20 FPS	Measure the update period
6	Audio frequency range	20 Hz - 20 kHz through audio signal path	Frequency sweep through the pedal and measure the output
7	Continuous operation	≥ 1 hour without crashing or audio loss	Operate the system continuously for one hour
8	<u>Effects library</u>	≥ 5 selectable digital effects	Demonstrate five different effects
9	<u>Simultaneous digital effects</u>	≥ 3 effects operating simultaneously	Enable maximum effects and visually and audibly confirm

Table 1. System Engineering Specifications Table

<i>Component Engineering Specifications</i>	
Audio Codec	
ADC/DAC Resolution	≥ 24 bits
Audio Sampling Rate	≥ 48 kHz
Number of Audio Channels	≥ 2 channels
ADC Dynamic Range	≥ 90 dB
DAC Dynamic Range	≥ 90 dB
Digital Audio Interface	I ² S compatible
Analog Supply Voltage	3.3 V or 5 V compatible
MCU	
CPU Cores	2 cores
Primary CPU Clock	≥ 100 MHz
On-Chip RAM	≥ 1 MB
Floating-Point Processing	Hardware FPU
Audio Communication Interface	≥ 1 I ² S/SAI interface
Peripheral Communication	SPI, I ² C, UART, GPIO, internal ADC
Supply Voltage	3.3 V or 5 V compatible
Signal Processing Capacity	≥ 2000 cycles/sample
Display	
Display Size	4.0–4.3 in diagonal
Display Resolution	$\geq 800 \times 480$ pixels
Display Type	IPS TFT LCD
Color Depth	≥ 16 -bit color
Refresh Rate	≥ 20 frames/s
Display Communication Interface	RGB / parallel interface
Operating Voltage	3.3 V or 5 V compatible
Power	
External Input Voltage	9–12 V DC
Digital Supply Rail	3.3 V $\pm$ 5%
Auxiliary Supply Rail	5.0 V $\pm$ 5%
3.3 V Rail Current Capacity	≥ 1.5 A
5 V Rail Current Capacity	≥ 1.0 A

Regulator Efficiency	$\geq 85\%$ at nominal load
Output Voltage Ripple	≤ 50 mV peak-to-peak

Table 2. Component Engineering Specifications Table

House of Quality

The House of Quality is a visual representation of the various goals of the product, the specifications that comprise those goals, and the tradeoffs between pursuing each goal. The marketing goals can be seen on the left, denoting what our product prioritizes in a broad sense. In the figure below (Figure 4), the highest priority is written in the 1) slot, second highest in 2), and so on.

The vertical section represents quantifiable specifications useful for determining how well the product's marketing priorities are being fulfilled. A "+" sign means that we seek to maximize the associated value, and conversely, a "-" sign means we seek the opposite. An up arrow means that the intersection between the two characteristics is positively correlated, e.g. maximizing sound quality and minimizing latency do not oppose one another. Furthermore, two upwards arrows indicate a strong positive correlation. Similarly, a single downwards arrow means a negative correlation, and two downwards arrows mean a strong negative correlation, e.g. minimizing CPU Temperature and maximizing Effects Count strongly oppose one another.

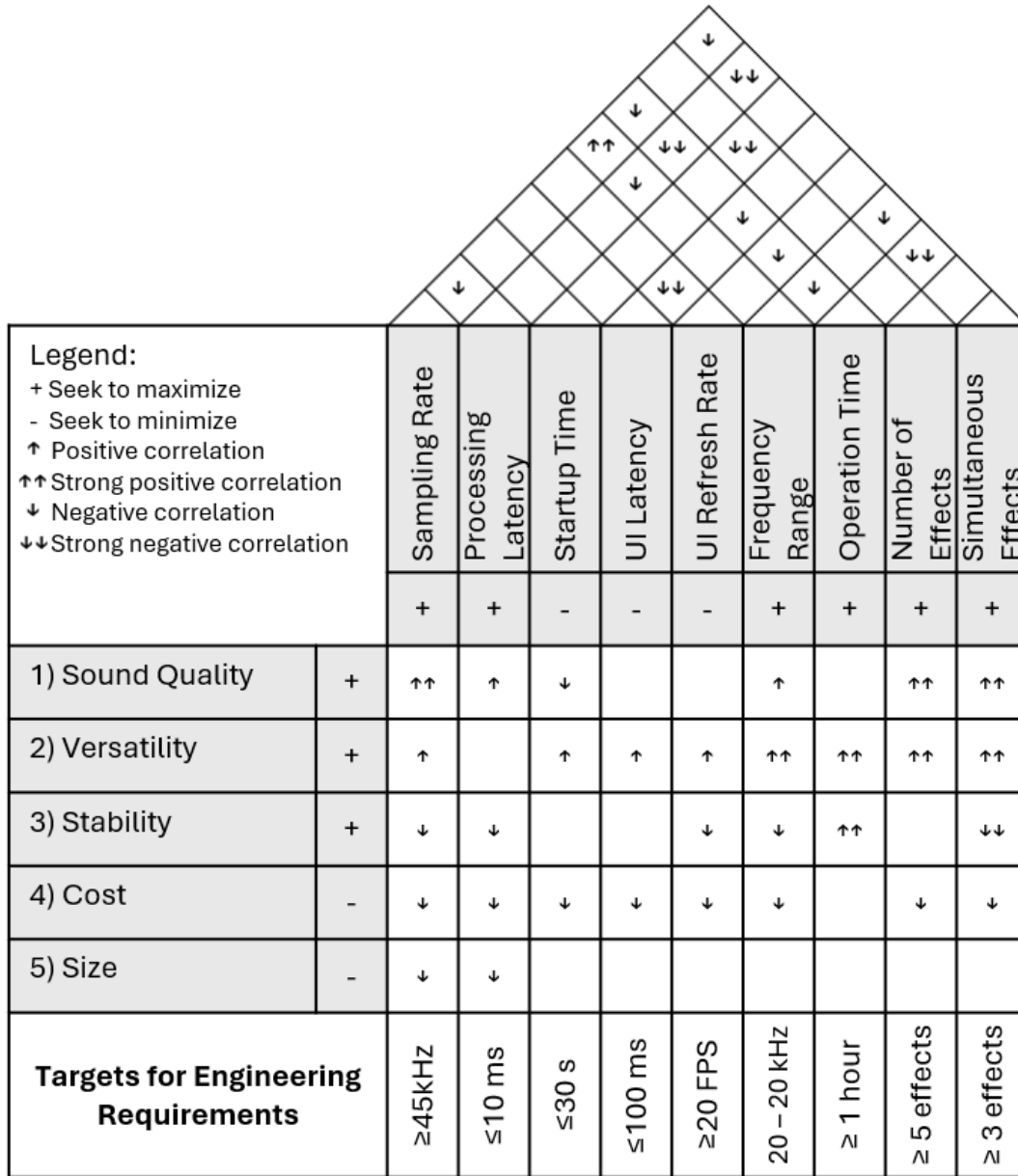


Figure 4. House of Quality

Hardware Block Diagram

A hardware block diagram is a useful overview of a system, providing a visual representation of major design choices as well as the interaction between components. In the below hardware block diagram (Figure 5), we seek to represent the multi-effects pedal as a general approach to solving the problem and achieving the marketing goals.

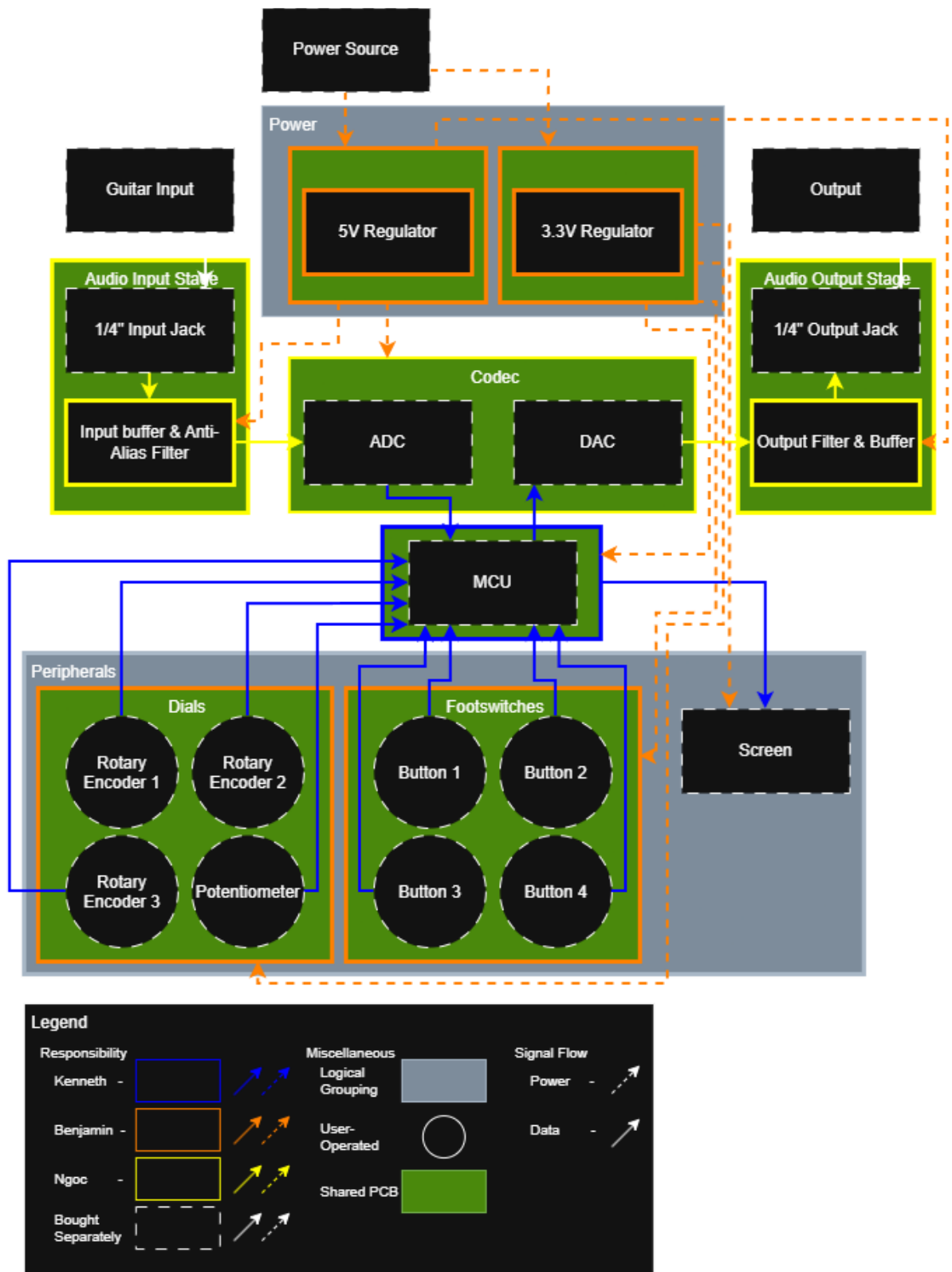


Figure 5. F.R.E.T. Hardware Block Diagram

To do this, we need to support a DSP device and allow the user to interact with the system. The user will be able to interact with input device(s), allowing control of the system, while the embedded hardware is supported by an input stage, output stage, and power subsystem.

Software Flowchart of Overall System

The user will load their own effects plugins, of which hundreds can be found online for free, to the single-board computer contained within the enclosure. Our software should allow the user to interact with their installed plugins via a screen, button(s), and rotary encoder(s).

To accomplish this, a software controller will be created or customized based on existing controllers, if suitable candidates can be found and appropriate permissions from the creators are obtained. The controller will allow the peripherals to act as an interface between the user and the plugins, whose operation is detailed below (Figure 6).

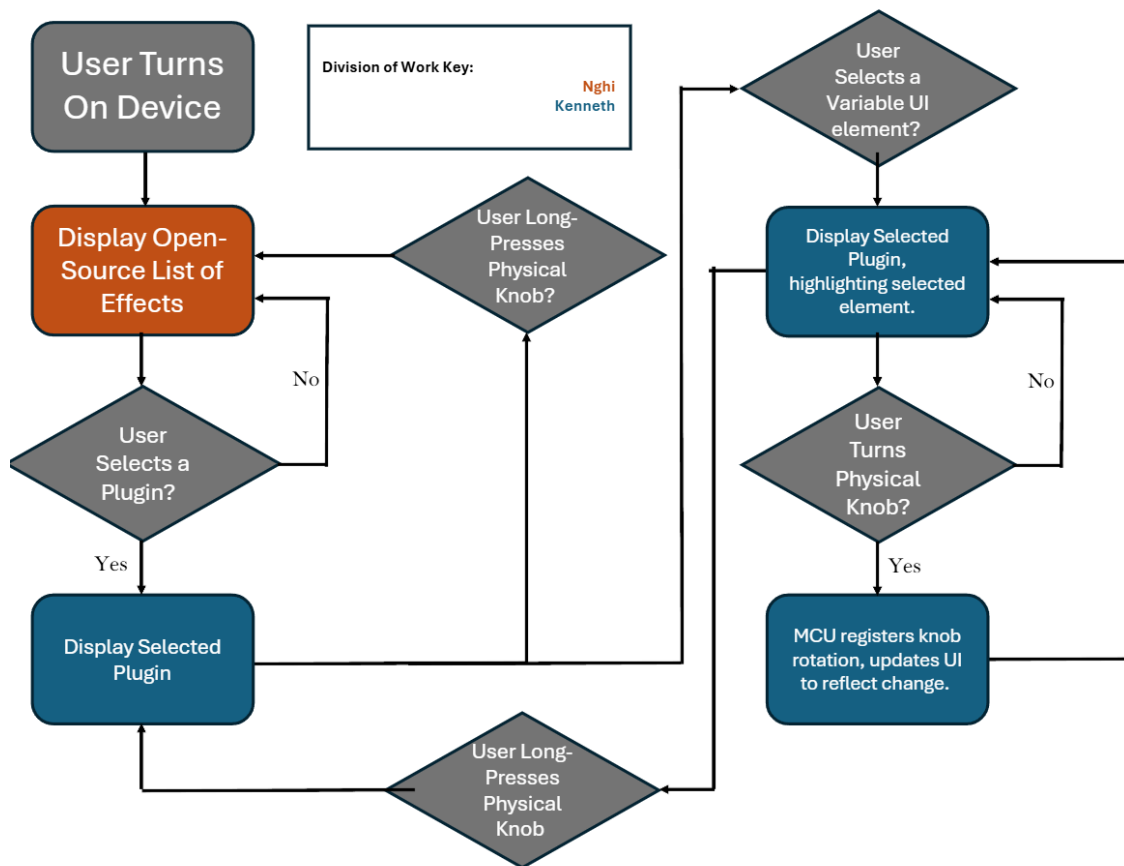


Figure 6. F.R.E.T. UI Software Block Diagram

The current design proposal involves utilizing a number of rotary encoders and buttons to operate the UI. The user may press the button to select a highlighted option, long press to go back, and rotate the encoder to move the highlighted option between neighboring UI elements or change the value of the element once selected (if applicable).

If all basic goals are met, an advanced goal is to monitor the temperature of the CPU itself, pictured in Figure 4. The MCU will, at regular intervals, determine the temperature of the CPU and activate an active cooling solution if a temperature threshold is exceeded.

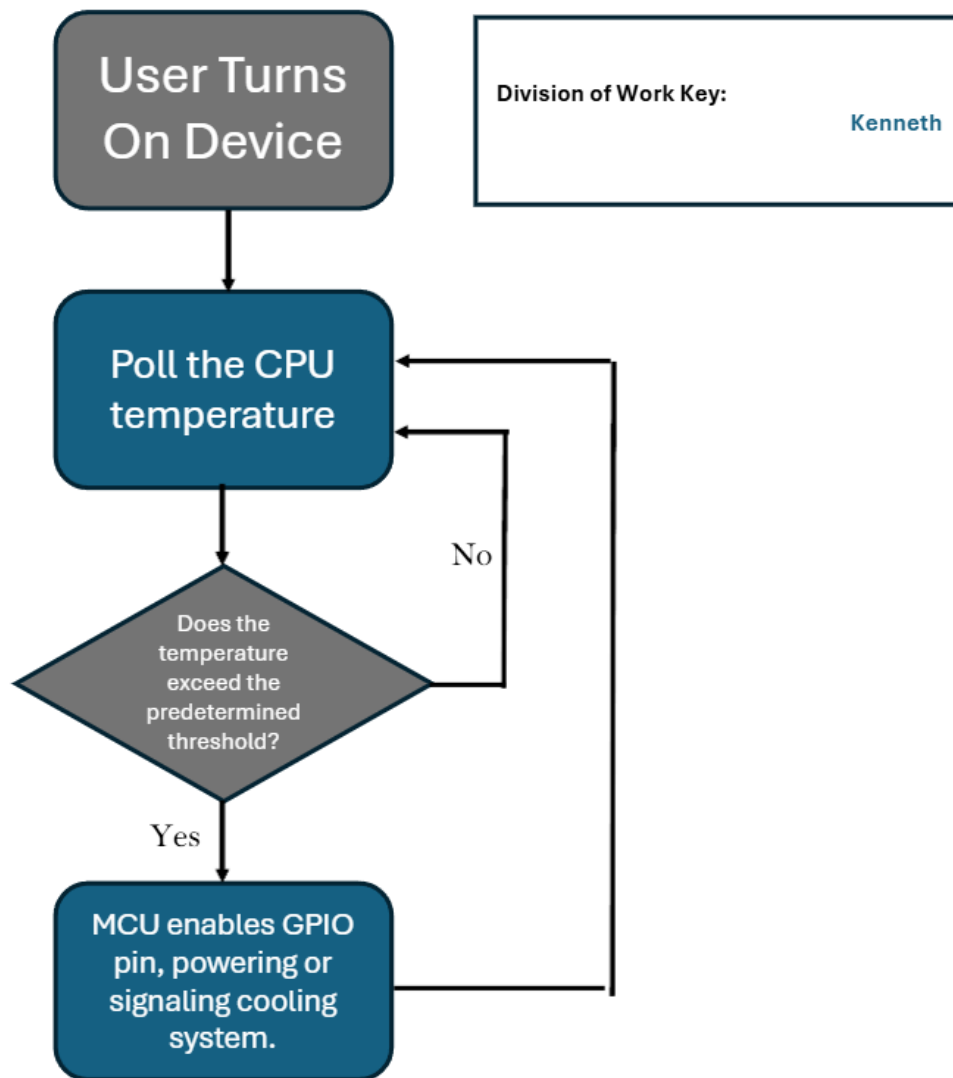


Figure 7. CPU Temperature Monitor Software Block Diagram

The poll will occur at a regular interval, depending on the needs of the selected CPU. Additionally, as a further goal, triggering the CPU temperature poll may be a result of enabling a plugin, detecting audio input, or some other trigger that causes or correlates with CPU activity.

Prototype Illustration/Blueprint

The blueprint (Figure 8) is built based on the design of the Pi-Stomp V3, with slight dimension changes to accommodate the larger display. The blueprint shows our general idea of what F.R.E.T should look like; this will most likely change in the future. One thing to note is that the spacing of the footswitches will stay relatively the same. They are 2.5 inches apart, for users to switch between effects without accidentally selecting a different open. Figure 9 shows the normal use case of F.R.E.T.

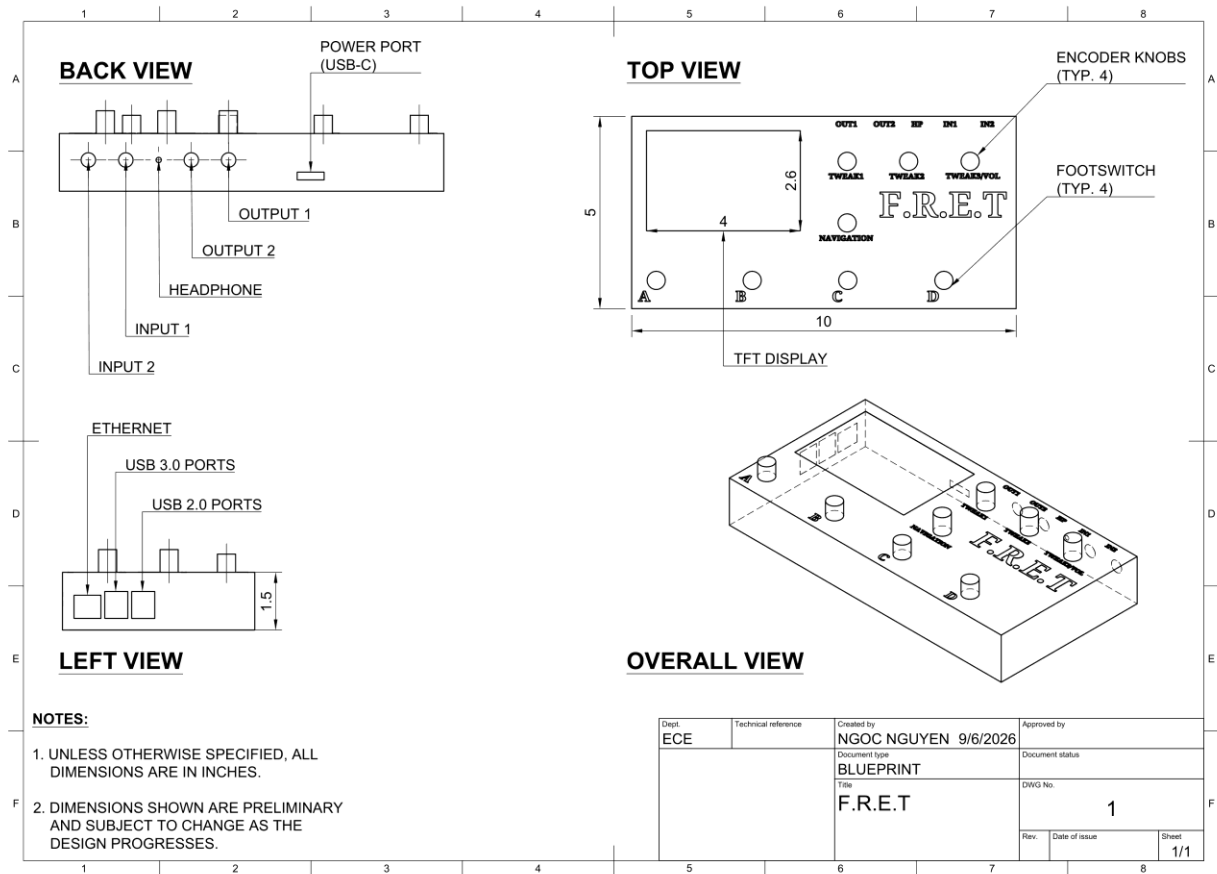


Figure 8. F.R.E.T. Tentative Blueprint

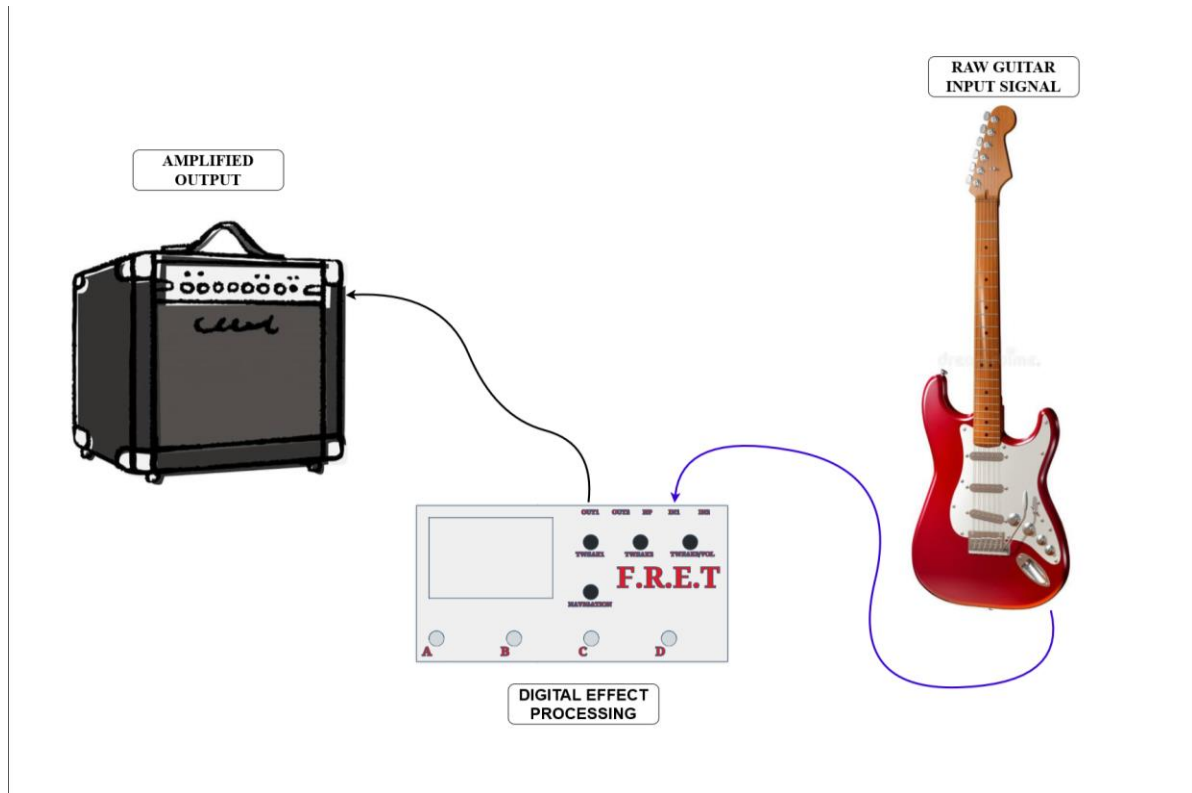


Figure 9. F.R.E.T. Illustration

Based on the illustration, the pedal stands between the guitar and the amplifier. Even though the guitar and amplifier are not in scope, they will be included to demonstrate the functions of the pedal.

Budget and Financing

The existing products mentioned are supplement guides for our project. The target is to build a hybrid product using good qualities of the existing products, while being able to improve certain aspects. We want to build a beginner friendly, affordable product without sacrificing quality. After taking everything into consideration, we have estimated the budget to be \$400.

Hardware	Budget
PCB	\$100
Rotary encoders and footswitch	

Project Milestones for SD1 and SD2

To remain on track, and provide structure to a long-term goal, we've come up with a tentative schedule of tasks which will help to facilitate timely completion of important milestones (Table 3).

Task	Dates	Responsibility
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Develop project proposal	9/4-9/25	Benjamin, Ngoc, Kenneth
Update Committee members	9/25-10/2	Benjamin, Ngoc, Kenneth
Prepare and design website for continued use	9/25-10/2	Kenneth
Initial parts acquisition	9/25-10/9	Benjamin, Ngoc, Kenneth
Design input and output buffer circuits	9/25-10/16	Benjamin
Begin writing embedded software	10/2-10/16	Kenneth, Ngoc
Design dev board MCU circuitry (peripherals, I/O)	10/2-10/16	Kenneth
Design power supply circuits	10/2-10/16	Ngoc
Update Committee and Website	10/9-10/16	Benjamin, Ngoc, Kenneth
Prototype and troubleshoot designed circuits in isolation. Compare strengths and weaknesses.	10/9-10/16	Benjamin, Ngoc, Kenneth
Implement prototyped circuits integrated with one another. Compare strengths and weaknesses.	10/16-10/23	Benjamin, Ngoc, Kenneth
Revise prototyped circuits, determine best implementation/design	10/16-10/23	Benjamin, Ngoc, Kenneth
Revise embedded software	10/16-10/23	Kenneth, Ngoc
Design standalone MCU board (move away from dev board)	10/16-10/23	Kenneth
Update Committee and Website	10/16-10/23	Benjamin, Ngoc, Kenneth
Write Midterm Report	10/16-11/6	Benjamin, Ngoc, Kenneth
Design and order buffer, power, MCU PCBs with components	10/23-11/6	Benjamin, Ngoc, Kenneth
Revise Midterm Report	11/6-11/13	Benjamin, Ngoc, Kenneth
Assemble, test, and troubleshoot PCBs	11/10-11/18	Benjamin, Ngoc, Kenneth
Update Website and Committee	11/13-11/20	Benjamin, Ngoc, Kenneth
Begin filming Mini Demo video	11/13-12/3	Benjamin, Ngoc, Kenneth
Begin Final Report	11/13-12/3	Benjamin, Ngoc, Kenneth
Revise PCB design and order updated version	11/13-11/18	Benjamin, Ngoc, Kenneth
Assemble PCBs, ver. 2	12/3-1/8	Benjamin, Ngoc, Kenneth
Senior Design 2 Begins	1/11	Benjamin, Ngoc, Kenneth

Update Website and Committee	1/11-1/15	Benjamin, Ngoc, Kenneth
Implement and Test Design	1/15-1/29	Benjamin, Ngoc, Kenneth
Major Revisions, or Begin Advanced and Stretch Goal designs	1/29-2/19	Benjamin, Ngoc, Kenneth
Update Website and Committee	1/29-2/5	Benjamin, Ngoc, Kenneth
Parts Acquisition	2/5-2/19	Benjamin, Ngoc, Kenneth
Update Website and Committee	2/19-2/26	Ngoc
Reattempt Basic goals, or Begin Advanced and Stretch Goal Prototypes	2/5-2/26	Benjamin, Ngoc, Kenneth
Design and order PCBs	2/26-3/5	Ngoc
Refine and polish design	3/19-4/16	Benjamin, Ngoc, Kenneth
Live Demo and Final Review Meeting	4/23-5/5	Benjamin, Ngoc, Kenneth

Table 3. Senior Design Milestones

Work Distribution

In order to facilitate the timely completion of F.R.E.T, objectives have been approximately divided into different sections and assigned to each member of the group, paying attention to previous experience, skill level, desire to learn a skill, and workload fairness. See Table 4 below for specifics.

Member	Task
Ngoc Nguyen	PCB Design
	Audio Input/Output Design
Benjamin Vargas	PSU Design
	Protective Circuitry, Risk Management
Kenneth Abel	MCU/Peripheral Design
	UI, DSP, Visualizer Software

Table 4. Work Distribution

Appendix A – Reference

- [A.1] Tree Fall Sound. “Pi-Stomp Manual.” *Pi-Stomp Manual - Pi-Stomp Manual*, treefallsound.github.io/pistomp-manual/home/. Accessed 11 Sept. 2026.
- [A.2] Sonicake. “SONICAKE QME-50 Matribox Multi-Effects Processor.” *Sonicake*, www.sonicake.com/products/matribox. Accessed 11 Sept. 2026.
- [A.3] Bloemer, Keith. “GuitarML/Funbox: Stereo Guitar Pedal Platform Using Daisy Seed.” *GitHub*, github.com/GuitarML/FunBox. Accessed 11 Sept. 2026.

Appendix B – Copyright and Usage Permission

Currently, no outside intellectual properties have been used in the creation of this document.

Appendix C – Miscellaneous